

7.2 Recommendations for System Improvement

1. **Implement ReAct-style feedback.** Add optional replanning after `run_cell` using scrape-based execution observers, closing the gap between dispatch-only Agentic mode and full autonomous repair (Shinn et al., 2023; Yao et al., 2023).
2. **Expand context packing.** Adopt hierarchical or retrieval-augmented cell selection for long notebooks to improve workflow-level Ask answers and Code placement (Lewis et al., 2020; Jiang et al., 2023).
3. **Harden DOM selectors.** Maintain scrape utilities against Kaggle front-end changes, following lessons from web data extraction research (Chasins et al., 2018).
4. **Add human evaluation.** Supplement automated rubrics with expert grading for explanation quality and code correctness (Chen et al., 2021; Wang et al., 2021).
5. **Live browser benchmarks.** Replicate harness scenarios on actual `/edit` sessions to measure end-to-end latency and user-perceived trust (Mozannar et al., 2024).

7.3 Recommendations for Future Research

- Compare GLM 4.7 with additional code models (Brown et al., 2020; OpenAI, 2023) on identical frozen snapshots.
- Study learning outcomes when Bachelor of Science in Data Science (BSDS) students use Ask versus Code versus Agentic assistance during coursework (Finnie-Ansley et al., 2022).
- Explore narrower tool APIs inspired by API-bank and Gorilla benchmarks (Li et al., 2023; Patil et al., 2023) tailored to competition notebooks (Quaranta et al., 2022).
- Investigate session-level memory across turns to reduce repeated query rounds in Agentic mode (Park et al., 2023).

These recommendations preserve the current system’s strengths: grounded context, mode separation, and bounded Agentic dispatch. They also identify clear paths toward fuller autonomous notebook assistance in future iterations.